

Practice 1: Ratio language

Name: _____ Date: _____

Write each ratio in the order named. Show your thinking.

1. A basket has 6 red apples and 4 green apples. What is the ratio of red apples to green apples?

Work space

2. There are 3 cats and 7 dogs. What is the ratio of dogs to all animals?

Work space

Practice 2: Equivalent ratios

Simplify only when it helps you compare. Show the multiplication or division.

1. Are the ratios 2:3 and 8:12 equivalent? Explain.

Work space

2. Complete the equivalent ratio: $5:2 = 15: \underline{\hspace{1cm}}$.

Work space

Practice 3: Ratio tables

Use a ratio table or another organized method.

1. One batch uses 2 cups of flour for every 3 cups of oats. How many cups of flour are needed for 12 cups of oats?

Work space

2. A team has 4 blue flags for every 5 yellow flags. Complete: $4:5 = \underline{\hspace{1cm}}:20$.

Work space

Practice 4: Ratios in pictures

Describe the relationship between the quantities.

1. A pattern has 9 triangles and 6 circles. Write the ratio triangles to circles in simplest form.

Work space

2. A shelf has 8 fiction books and 12 nonfiction books. What fraction of the books are fiction? Write the matching ratio of fiction to all books.

Work space

Practice 5: Comparing ratios

Use equivalent ratios or a table to justify each answer.

1. Which has more blue per red: 6 blue and 4 red, or 9 blue and 7 red? Explain.

Work space

2. Are 3:4 and 12:16 equal ratios? Give one reason.

Work space

Practice 6: Ratio word problems

Identify both quantities before calculating.

1. For every 2 laps Maya runs, she walks 3 laps. If she runs 8 laps, how many laps does she walk?

Work space

2. A recipe uses 3 lemons for 4 pitchers. How many lemons are needed for 12 pitchers?

Work space

Practice 7: Unit rates

Find the amount for one unit.

1. Six notebooks cost \$18. What is the cost per notebook?

Work space

2. A car travels 240 miles in 4 hours. What is its rate in miles per hour?

Work space

Practice 8: Rate tables

Complete the table using a constant rate.

1. At 5 pages per minute, how many pages can be read in 3, 7, and 10 minutes?

Work space

2. Four tickets cost \$28. At the same rate, what do 9 tickets cost?

Work space

Practice 9: Unit-rate comparisons

Compare the per-item or per-unit amounts.

1. Store A sells 3 pens for \$6. Store B sells 5 pens for \$12. Which store has the lower unit price?

Work space

2. A runner goes 2 miles in 18 minutes. At the same pace, how long for 5 miles?

Work space

Practice 10: Measurement rates

Include units in every answer.

1. A faucet fills 12 liters in 3 minutes. How many liters per minute?

Work space

2. A printer makes 45 pages in 5 minutes. How many pages in 8 minutes at that rate?

Work space

Practice 11: Percent means per hundred

Write each percent as a fraction out of 100 and as a decimal.

1. Write 35% as a fraction with denominator 100 and as a decimal.

Work space

2. What percent is represented by 0.62? Write it as a fraction out of 100.

Work space

Practice 12: Percent models

Use a 10-by-10 grid idea or fraction reasoning.

1. A class has 100 squares in a chart and 48 are shaded. What percent is shaded?

Work space

2. Find 25% of 80. Explain using quarters.

Work space

Practice 13: Percent of a whole

Find the percent of the given number.

1. Find 10% of 340.

Work space

2. Find 50% of 126. Explain your strategy.

Work space

Practice 14: Common benchmark percents

Use 10%, 25%, 50%, or 75% benchmarks.

1. Find 75% of 60.

Work space

2. Find 20% of 250.

Work space

Practice 15: Missing percent

Find the unknown percent or whole.

1. 30 is what percent of 120?

Work space

2. 40% of what number is 28?

Work space

Practice 16: Percent word problems

Underline the whole before solving.

1. A library has 200 books. Twenty-five percent are biographies. How many biographies are there?

Work space

2. A garden has 80 plants; 35% are tomatoes. How many plants are not tomatoes?

Work space

Practice 17: Discounts

Find the discount, then the sale price.

1. A \$40 backpack is 25% off. What is the discount and sale price?

Work space

2. A \$60 game is 10% off. What is the sale price?

Work space

Practice 18: Tax and totals

Use the stated whole-number percent.

1. A \$50 purchase has 8% tax. How much is the tax and total cost?

Work space

2. A \$25 meal has 20% tip. What is the tip and total?

Work space

Practice 19: Mixed review

Choose a ratio, rate, or percent method and explain why it fits.

1. A map uses 1 inch for every 4 miles. How many miles are represented by 6 inches?

Work space

2. A team won 18 of 24 games. What percent of its games did it win?

Work space

Practice 20: Final challenge

Show organized work and label every answer.

1. A school orders 240 pencils. They are packed in boxes with 12 pencils each. If 25% of the boxes are blue, how many boxes are blue?

Work space

2. A cyclist travels 36 miles in 3 hours. At that rate, how far in 5 hours, and what percent of 100 miles is that distance?

Work space